

Our Environment

Ecosystem and its parts, food chains, food webs and energy flow, ozone layer and its protection, and how we manage biodegradable and non biodegradable waste.

Environment and Ecosystem

Environment

Environment is everything that surrounds an organism and influences its life. It includes physical factors like air, water, soil, light and temperature, and biological factors like plants, animals and microbes.

Ecosystem

An ecosystem is a system formed by the interaction of all living organisms with each other and with the physical and chemical factors of the environment. The term was given by A. G. Tansley in 1935.

An ecosystem includes both biotic components, the living parts, and abiotic components, the non living parts. All parts interact and are linked by energy flow and nutrient cycles.

Examples: A pond, a lake, a forest, a grassland, a desert, a field and even an aquarium can be considered as ecosystems. Some are natural ecosystems like forests. Some are human made or artificial ecosystems like a crop field, a garden and an aquarium.

An aquarium is an artificial and almost complete ecosystem because it has plants, fish and microbes, but it needs regular cleaning and supply of food and oxygen from outside, so it is not fully self sustaining.

Components of Ecosystem

Biotic Components

Biotic components are the living parts of the ecosystem. They are of three types:

- **Producers:** All green plants and some bacteria that can make their own food by photosynthesis. They produce organic matter from inorganic substances using sunlight. They are also called autotrophs.
- **Consumers:** Organisms that cannot make their own food and depend on producers. They are also called heterotrophs. Consumers are of different types: herbivores that eat plants, carnivores that eat animals, omnivores that eat both plants and animals, and parasites that live on a host.
- **Decomposers:** Bacteria and fungi that break down dead plants and animals into simple inorganic substances. They clean the environment and return nutrients to the soil and air for reuse by producers. Without decomposers, nutrients would not be recycled.

Abiotic Components

Abiotic components are the non living physical and chemical factors:

- **Physical factors:** Sunlight, temperature, rainfall, wind, soil and water.
- **Chemical factors:** Oxygen, carbon dioxide, nitrogen, minerals, pH and salinity of soil and water.

Energy from the sun flows through the biotic components, and materials are cycled between biotic and abiotic parts with the help of decomposers.

Functions of an Ecosystem

An ecosystem carries out two main functions: flow of energy and cycling of nutrients. Energy enters as sunlight, is captured by producers and moves through consumers step by step. Nutrients like carbon, nitrogen and water cycle between living and non living parts.

Food Chains and Food Webs

Food chain

A food chain is a sequence of organisms where each organism eats the one below it and is eaten by the one above it. It shows the transfer of food energy step by step.

Example of a food chain in a grassland:

Grass → Deer → Tiger → Decomposers

Or in a pond:

Algae → Protozoa → Small fish → Large fish

A more common example given in NCERT:

Grass → Insect → Frog → Snake → Eagle

Food Web

Food web

A food web is a network of interconnected food chains. In nature, one organism can be eaten by many organisms and can eat many organisms. So the chains are linked together.

In a food web, if one organism is removed, other paths can still supply food, so the web is more stable than a single chain. But if a key organism is lost, the whole web is affected.

Trophic Levels and Energy Flow

Trophic level

Trophic level is the position or step that an organism occupies in a food chain. The levels show who feeds on whom.

- **First trophic level:** Producers, green plants and algae.
- **Second trophic level:** Primary consumers, herbivores like deer, insects and zooplankton.
- **Third trophic level:** Secondary consumers, small carnivores like frog and small fish.
- **Fourth trophic level:** Tertiary consumers, larger carnivores like snake and large fish.
- **Fifth trophic level:** Top carnivores like eagle, tiger and lion. Decomposers act at all levels.

Each trophic level transfers energy to the next level, but only a small part of the energy is transferred.

10 Percent Law

10 Percent Law

Only about 10 percent of the energy available at one trophic level is transferred to the next trophic level. The remaining 90 percent is lost as heat, respiration, incomplete digestion and other life processes.

This was given by Raymond Lindeman. So if producers have 10,000 J of energy, primary consumers get about 1,000 J, secondary consumers get about 100 J and tertiary consumers get about 10 J.

Because progressively less energy is available at higher trophic levels, food chains are generally short—often three or four trophic levels. The organisms at the highest level receive the least share of the original energy captured by producers.

Worked Example

If a plant has 5000 J of energy, how much energy will be available to the snake in the chain: Plant → Insect → Frog → Snake?

Plant, T1 = 5000 J → Insect, T2 = 10% of 5000 = 500 J → Frog, T3 = 10% of 500 = 50 J → Snake, T4 = 10% of 50 = 5 J

Only 5 J reaches the snake.

Energy flow in an ecosystem is always unidirectional, from producers to consumers. It does not cycle back. Nutrients, on the other hand, are recycled.

Why are some substances not biodegradable

Some substances made by humans, like plastics, do not break down by biological processes, so they remain in the environment for a long time.

Bioaccumulation and Biomagnification

Harmful chemicals like pesticides and heavy metals enter the food chain through water and soil. They are not broken down inside the body and they accumulate.

Biomagnification

Biomagnification is the increase in concentration of harmful chemicals at each successive trophic level of a food chain. The highest level gets the maximum concentration.

Example: Pesticides sprayed on crops enter soil and water, are taken up by plants, then by insects, then by birds that eat insects, then by humans. At the top level, the concentration can be very high and cause health problems like cancer and reproductive issues. Pesticides like DDT showed strong biomagnification.

This is why the use of non biodegradable pesticides should be reduced and behaviour like biological magnification should be studied before approving a new chemical.

Ozone Layer and its Depletion

Ozone layer

Ozone is a molecule made of three oxygen atoms, O₃. The ozone layer is a layer in the stratosphere, about 15 to 30 km above the earth, that contains a high concentration of ozone.

Formation: Ozone is formed when ultraviolet rays split oxygen molecules, O₂, into single oxygen atoms, O. These atoms then combine with O₂ to form O₃. **Destruction:** O₃ is also broken by UV rays into O₂ and O. So a balance is maintained.



Function: The ozone layer absorbs most of the harmful ultraviolet rays from the sun and protects living organisms. UV rays can cause skin cancer, eye damage and harm to crops and marine life.

Ozone Depletion

In the 1980s, scientists found that the amount of ozone was dropping, especially over Antarctica. This thinning is called ozone depletion and the thin area is called the ozone hole.

Cause: The main cause is chlorofluorocarbons, CFCs. These are chemicals used in refrigerators, air conditioners, aerosol sprays and foam packaging. CFCs release chlorine atoms in the stratosphere. One chlorine atom can destroy many ozone molecules repeatedly.

Effect: More UV rays reach the earth, increasing the risk of skin cancer, cataracts, damage to the immune system and harm to plants and marine food chains.

Solution: In 1987, the United Nations Environment Programme, UNEP, brought nations together to sign the Montreal Protocol. Countries agreed to freeze and reduce the production and use of CFCs and other ozone depleting substances. As a result, ozone depletion has slowed and recovery is expected over decades.

In NCERT, the role of UNEP in the Montreal Protocol and the ban on CFCs is highlighted as a successful example of international cooperation.

Managing Garbage and Waste

Waste and garbage are produced every day in homes, schools, factories and hospitals. Managing them properly is essential to keep the environment clean.

Biodegradable and non biodegradable waste

Biodegradable waste can be broken down by biological processes with the help of microbes like bacteria and fungi. Examples are vegetable peels, paper, cotton and animal waste.

Non biodegradable waste cannot be broken down by biological processes. It stays in the environment for a long time and harms it. Examples are plastic, glass, metal and synthetic chemicals.

Differences:

- Biodegradable waste decomposes quickly, does not accumulate much, and can be composted or used for biogas. But it can produce foul smell and breed germs if left open.
- Non biodegradable waste persists for years, accumulates, can block drains, harm animals and enter food chains. It needs to be reduced, reused and recycled.

Methods of Waste Management

- **Reduce:** Produce less waste by avoiding single use plastics and over packaging.
- **Reuse:** Use bottles, bags and containers again instead of throwing them away.
- **Recycle:** Collect paper, glass, metal and some plastics and make new products.
- **Composting:** Convert biodegradable waste into manure for fields.
- **Biogas:** Use cow dung and plant waste in biogas plants to produce fuel.
- **Sewage treatment:** Treat liquid waste before releasing into rivers.
- **Sanitary landfill:** Dispose of solid waste in low lying areas away from cities with proper lining and covering.

Our habits: Use cloth bags instead of polythene bags, separate biodegradable and non biodegradable waste at source, repair and reuse items, and say no to single use plastic. Improved practices like using kulhads instead of plastic cups were studied, but even paper cups have an impact because trees are cut to make paper, so reducing use is the best option.

Summary

- Environment includes living and non living surroundings. Ecosystem is the interaction of biotic and abiotic components for energy flow and nutrient cycling. It can be natural or artificial.
- Biotic parts are producers, consumers and decomposers. Abiotic parts are physical and chemical factors like light, temperature, water and nutrients.
- Food chain is a linear sequence of feeding. Food web is a network of linked chains. Trophic level is the position in the chain.
- Only 10 percent of energy moves to the next trophic level. Energy flow is unidirectional. Food chains have 3 to 4 levels because little energy remains at the top.
- Harmful chemicals show biomagnification, their concentration increases at higher trophic levels.
- Ozone, O₃, in the stratosphere absorbs UV rays. CFCs from refrigerators and sprays release chlorine and deplete ozone, causing the ozone hole. The Montreal Protocol of UNEP in 1987 controls CFCs.
- Waste is biodegradable, broken down by microbes, or non biodegradable, not broken down. Management is by reduce, reuse, recycle, composting, biogas and proper disposal.

Practice Questions

1. What is an ecosystem? Distinguish between natural and artificial ecosystems. Give two examples of each.

2. Describe the biotic and abiotic components of an ecosystem. What is the role of decomposers?

3. Differentiate between food chain and food web. Draw a food chain with four trophic levels found in a grassland.

4. State the 10 percent law. If 10,000 J of energy is available at the producer level, how much energy will be available to the tiger through the chain: Plant → Deer → Tiger?

5. What is biomagnification? Explain how pesticides reach humans through a food chain.

6. What is ozone and why is the ozone layer important? Name the chemical that depletes ozone and the international agreement made to protect it.
Differentiate between biodegradable and non biodegradable waste with examples.